



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.:7

Team No.: 17

Project Title: Kernel Monitor

Team Members

Roll Number	Student Name	Signature
2520030322	Challagulla Shravya	
2520030101	Gandu Rishitha	
2520030509	M.V.L.Samhita	

1. Abstract

The **Kernel Monitor** is a Linux-based system monitoring application designed to provide real-time information about important operating system activities. Modern systems run multiple processes and background operations simultaneously, making it difficult to identify high resource usage, abnormal processes, and performance issues. The proposed system monitors parameters such as **CPU utilization, memory usage, running processes, process states, and system resource consumption** using Linux system interfaces and system programming techniques. It collects and processes system-level information and presents it through a simple and user-friendly monitoring interface. The project demonstrates important Operating Systems concepts including **process management, memory management, system calls, file systems, and resource monitoring**. The system can also identify excessive resource utilization and generate alerts when predefined limits are crossed. The expected outcome is a practical Linux-based monitoring tool that provides better visibility into system behavior and helps users identify performance-related issues efficiently.

2. Problem Statement

Modern operating systems execute numerous processes and system operations simultaneously. Identifying resource-intensive processes, excessive CPU or memory usage, and abnormal system activity can be difficult using basic system information. Therefore, a Linux-based **Kernel Monitor** is required to continuously monitor important system and process-level activities and present them in an understandable format. The project focuses on applying Operating Systems and System Programming concepts to monitor system performance and resource utilization in real time.

3. Objectives

1. To monitor **CPU, memory, and system resource utilization** in real time.
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2. To track **running processes and their resource consumption**.

3. To collect system-level information using **Linux interfaces and system calls**.

4. To identify excessive resource usage and provide **useful alerts and monitoring information**.

4. Proposed Methodology

The system will be developed on **Linux using C/C++**. It will collect system and process information from Linux kernel interfaces such as the /proc and /sys filesystems. The collected information will be processed to calculate CPU usage, memory utilization, process statistics, and other system parameters. The monitor will periodically refresh the collected data and display it through a simple interface. Threshold-based monitoring will be implemented to identify unusually high resource consumption. The overall flow will be **data collection → processing → analysis → display → alert generation**.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Process Management	Monitor running processes and process states
Memory Management	Track RAM and memory utilization
/proc Filesystem	Obtain process and system information
/sys Filesystem	Access system and hardware-related information
fork()	Demonstrate/process creation where required
getpid()	Obtain process identification
open(), read(), close()	Read system information from Linux interfaces
Signals	Handle monitoring events and alerts
POSIX APIs	Perform system-level operations

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030322	Challagulla Shravya	Monitor running processes, process IDs, process states, CPU usage per process; use getpid(), fork(), and read process information from /proc.
2520030101	Gandu Rishitha	Monitor overall CPU utilization, RAM/memory usage, system resource consumption; use /proc and /sys to obtain system/hardware information.
2520030509	M.V.L.Samhita	Implement open(), read(), close() POSIX APIs and signals; process collected data and create the monitoring interface + alerts when resource limits are exceeded.

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Development and execution environment
C / C++	System-level programming
GCC Compiler	Compiling the source code
Linux /proc and /sys	Collecting system information
Git & GitHub	Version control and project repository
VS Code / Vim	Code development and editing

8. Expected Outcome

The project is expected to provide a **working Linux-based Kernel Monitor** capable of displaying real-time CPU usage, memory utilization, process information, and system resource consumption. It will help users identify resource-intensive processes and abnormal resource usage through monitoring and alerts. The project will also demonstrate practical understanding of **process management, memory management, Linux system interfaces, system calls, and system programming**.

9. GitHub Repository

GitHub Repository Link: _____

Repository Created: _____

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: _____

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____